Tutorial 07 — Testing: Report

Software Development Principles and Practice, 2026 Fall · Spec-driven testing of the due-date feature

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Forked repository	https://github.com/anyutkinz/SWPP-2026-Testing-Practice
Pull request (in the fork)	https://github.com/anyutkinz/SWPP-2026-Testing-Practice/pull/1 — `feature/due-date` → `main`
CI run	https://github.com/anyutkinz/SWPP-2026-Testing-Practice/actions/runs/39366136282 Workflow "Android Tests" on commit `75ee160` (event `pull_request`, 2026-10-20 03:47 UTC). Conclusion **failure**, but **no test executed**: both jobs (`unit-test`, `integration-test`) ended after about 2 s without a runner or any step, with the annotation "The job was not started because your account is locked due to a billing issue."
Tested commit	`26220c868b4dbf0ea242ad5485b01ea216e3ef39` (specs + tests; production code identical to `7e091a7`)
Target functions	`dueDateLabel()` in `tasks/DueDateUtils.kt`; `setDueDateLabel()` in `tasks/TasksListBindings.kt`

1. Summary

- **Bug 1 (wrong D-day count) — detected.** 11 of 34 unit tests and 1 instrumented binding test fail with the day-of-month value instead of the total day count, e.g. 2026-10-20 → 2026-12-01 gives `D-11` instead of `D-42`.
- **Bug 2 (stale label not cleared) — detected.** 3 direct-call binding tests that reuse one `TextView`, plus the supplementary screen test, fail: after unchecking a task without a due date the row still shows a visible `CLEAR`.
- **CI:** the PR triggered the workflow, but GitHub refused to start both jobs because the GitHub account is locked for a billing issue, so **no test ran in CI**. All results below are from local runs (API 37 and API 30 emulators).
- All other new tests pass. Every failure is an assertion failure; there were no compilation errors, test errors or skipped tests in the recorded runs.
- Both bugs were already exposed by scenarios present in the *AI-generated* specs. The human-reviewed refinements made the oracle more precise and added boundary and traceability detail; they were not needed to detect either bug (details in §4 and §7).

2. Process

1. **Inspection.** The coding agent (Claude Code, model Claude Opus 5.5) read the assignment PDF, `AGENTS.md`, `specs/TEMPLATE.spec.md`, the two target functions, the related model/layout/adapter code, the existing tests, the Gradle configuration, the CI workflow and the original PR description. The initial feature commit was recorded as `7e091a7d933507fe08d07c2d301310b252351d37`.
2. **AI specs.** Following `AGENTS.md`, the agent copied the template to `specs/DueDateUtils.spec.md` and `specs/TasksListBindings.spec.md`, filled them in from the feature intent (slide 53 and the original PR description), and saved unchanged copies as `DueDateUtils-AI.spec.md` and `TasksListBindings-AI.spec.md` (commit `e517122`). Expected calendar values were computed independently with Python `datetime` rather than `java.time.Period`.

   > **Disclosure.** The agent drafted the "AI" specs with the whole PDF available, including slide 68, which names the two bugs. The drafts are therefore not "blind" with respect to the bugs. The agent was instructed not to omit cases or plant mistakes to create a refinement story, and the drafts were preserved exactly as first written.

3. **Human review.** The agent critically reviewed its drafts against the assignment checklist and presented seven proposed refinements (R1–R7), together with the sections it considered already correct. The student reviewed this checkpoint summary and these proposals, approved R1–R6, approved R7 as supplementary screen-level coverage outside the spec template, chose the "no label" definition, added reproducibility requirements for date-

relative fixtures, and asked for the midnight undefined-behaviour statement to be replaced (R8). The student made these decisions on the agent's summary; the student did not independently inspect every specification line, and the wording of the changes was written by the agent.

4. **Refined specs.** The agent applied the decisions (commit `df03528`) before writing any test.
5. **Tests.** One test per Tip was written from the refined specs (commit `26220c8`): 34 unit tests in `app/src/test` and 16 instrumented tests in `app/src/androidTest`, named `subject_actionOrInput_result` with Given/When/Then, plus one supplementary screen-level test. No production code was changed.
6. **Execution.** `testDebugUnitTest` and `connectedDebugAndroidTest` were run locally: first with the final test files before they were committed, then on a clean checkout of `26220c8` (API 37 emulator), and the instrumented suite once more on an API 30 emulator. The work was pushed to the PR in the fork; CI was triggered but GitHub did not start the jobs (§6.4).

3. Specifications

`dueDateLabel(dueDate: LocalDate?, today: LocalDate, isCompleted: Boolean): String` is a pure function: completed → `"CLEAR"`; active without a due date → `""` (the "no label" value, as documented in its KDoc and implied by the non-null return type); otherwise `"D-N"`, `"D-Day"` or `"D+N"`, where N is the total number of calendar days. All its post-conditions are `[Unit]`.

`setDueDateLabel(textView: TextView, task: Task?)` reads `LocalDate.now()` itself and writes the label into a `TextView`. It needs the Android runtime (the project has no Robolectric), so all its post-conditions are `[Integration]`. "Shows X" means exact text X *and* `View.VISIBLE`; "shows no label" means `textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE` (option (a), chosen by the student; neither the KDoc nor the assignment requires a stronger "empty and GONE" contract). A visible stale `CLEAR` therefore fails.

Final specs: 34 Tips for `dueDateLabel()` (AI draft: 31) and 16 Tips for `setDueDateLabel()` (AI draft: 16). Neither spec uses `Tested by`, because no pre-existing test checked these functions; the template structure is unchanged.

4. Specification refinements (AI draft → final)

"Before" and "After" quote the files; the tests named are the ones that check the refined condition. The `-AI` files are byte-for-byte unchanged (SHA-256 in §9).

ID	Location	Before (AI draft)	After (final)	Reason / requirement	Test(s)
R1	DueDateUtils → Pre-conditions, 1st bullet	"`today` is the current local date, supplied by the caller (the function does not read the clock)."	"`today` is any `LocalDate`; it is the reference date the label is computed against. The app passes the device's current date, but the function never reads the clock itself, so any fixed date (for example 2028-02-28) is valid input."	A pre-condition must not exclude ordinary valid input. Read literally, the draft made every fixed-date Tip (2027-01-01, 2028-02-28, …) violate the pre-condition. Keeps unit tests deterministic.	No new test; legitimises all 34 fixed-date tests, e.g. `dueDateLabel_dueLeapDayFromFeb28_returnsDMinus1`.
R2	DueDateUtils → Edge cases, "Year end"	"Year end, one day apart in each direction." Two Tips: today 2026-12-31 → due 2027-01-01 = `D-1`; today 2027-01-01 → due 2026-12-31 = `D+1`.	"Year end: `today` is the last day of the year and the due date is one day before it, on it and one day after it; plus the reverse direction…" Added: due 2026-12-30 → `D+1`; due 2026-12-31 → `D-Day` (today 2026-12-31). Both existing Tips kept.	`AGENTS.md`: edge cases at the boundary and right next to it on both sides. The draft had only the two crossing points; the "on" point and the same-year side were missing (the month-end group already had all three). The reverse-direction Tip is kept because it crosses the year end in the overdue direction.	`dueDateLabel_dueDayBeforeYearEnd_returnsDPlus1`, `dueDateLabel_dueOnYearEnd_returnsDDay` (both pass; one-day intervals cannot show bug 1).

ID	Location	Before (AI draft)	After (final)	Reason / requirement	Test(s)
R3	DueDateUtils → Success cases (new)	No overdue case that both crosses a year end and is longer than a month (the future direction had `D-52`).	"Active task overdue across a year end by more than one month…" Tip: dueDate 2026-11-20, today 2027-01-10 → `"D+51"` (51 days, verified independently).	Symmetry with the future-direction case; combines year boundary and long interval in the overdue direction.	`dueDateLabel_activeOverdueBy51DaysAcrossYearEnd_returnsDPlus51` — fails (`D+20`): extra evidence for bug 1, not needed to detect it.
R4	TasksListBindings → Pre-conditions, 4th bullet → Post-conditions	Pre-condition: "'Shows no label' means the `TextView` displays no text: its text is `""` or its visibility is not `View.VISIBLE`." Tips: "Then textView.text is "" or textView.visibility is not View.VISIBLE."	Defined in the post-condition "Active task with no due date: … shows no label, i.e. `textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE` is true. Every 'shows no label' below means exactly this; a visible non-empty label such as a stale "CLEAR" is not 'no label'." All no-label Tips use the same expression.	The definition is part of the oracle, not something that must hold before the call. The substance is the same as the draft (option (a), chosen by the student); the change makes it a single, exact, consistently referenced observable.	`setDueDateLabel_activeNoDueDate_showsNoLabel`, … `_nullTask_showsNoLabel`, … `_noDueDateTaskCheckedThenUnchecked_…`, … `_labelledViewReusedForTaskWithoutLabel_…`, … `_clearViewBoundToNull_…`.
R5	TasksListBindings → Pre-conditions, 3rd bullet + every date-dependent Tip + 42-day condition	"'Today' is the device's current local date (`LocalDate.now()`); a test builds its due dates from that same date and does not run across midnight." Tips used `LocalDate.now().plusDays(3)`, `.minusDays(3)`, … without examples.	Explicit recipe: `today = LocalDate.now()` read on the device under test (not the host), `dueDate = today.plusDays(k)`; the expected literal depends only on k. Examples: 2026-10-20 + 3 → 2026-10-23 → `"D-3"`; + 42 → 2026-12-01 → `"D-42"`; − 3 → 2026-10-17 → `"D+3"`. Each Tip states k and an example date. 42-day condition adds: "42 days always span more than one calendar month, so "D-42" separates the total day count from a day-of-month difference on every run date." New bullet: a device-date change during an attempt is a test-environment race; the attempt is discarded and repeated with regenerated fixtures and never asserted.	`AGENTS.md` asks for exact input dates, but the function reads the clock and production code may not change, so absolute dates are impossible. The adaptation keeps the expected strings exact and makes the input recipe reproducible. The student added the requirements to use the device date, record the actual dates, and handle the midnight race explicitly.	All date-dependent binding tests; helper `onStableDeviceDate` logs the reference date, due date and attempt number (§6.3).
R6	TasksListBindings → Edge −1/0/+1 Tips, 42-day Tip, both check/uncheck Tips	These Tips checked only `textView.text` (the toggle Tips checked visibility only after the last call or not at all).	"… and `textView.visibility` is `View.VISIBLE`" added; the check/uncheck-with-due-date Tip checks VISIBLE "after each call", and the no-due-date one checks it after the "CLEAR" call.	A label is shown only if it is visible; the other label Tips already checked both. Makes the "shows X" definition apply uniformly.	… `_activeDueYesterday_showsDPlus1`, … `_activeDueOnTodayBoundary_showsDDay`, … `_activeDueTomorrow_showsDMinus1`, … `_activeDue42DaysAhead_showsDMinus42`, both check/uncheck tests.

ID	Location	Before (AI draft)	After (final)	Reason / requirement	Test(s)
R7	Not in the spec (report test plan only)	The check/uncheck intent was covered only by direct calls that reuse one `TextView`.	Supplementary instrumented test `TasksFragmentDueDateTest.tasksList_checkThenUncheckTaskWithoutDueDate_showsClearThenNoLabel`, mapped to the edge case "same `TextView` bound again after an active task with no due date is checked and then unchecked" (TLB-12).	Slide 53: "the label should be updated when checking or unchecking a completed task." Shows the defect through the real list row, data binding and checkbox. Per the student's decision, the spec keeps the template's direct-call contract and is not restructured.	The screen test (§6.3). The direct-call tests are kept as independent evidence.
R8	TasksListBindings → Undefined behavior	"A call that runs across midnight: `LocalDate.now()` can change between building the test's due date and the call, so the label may be off by one day. Tests must not depend on that moment."	"None. Every input has a defined result; a device date change during a test is handled as a test-environment race (see Pre-conditions), not as application behavior."	Student decision: a fixture crossing midnight is a property of the test, not undefined behaviour of the application. It is now handled explicitly (R5).	Implemented in `onStableDeviceDate` (bounded to 3 attempts; assertions only on a stable attempt).

Sections reviewed and kept unchanged: both Intents; "Error cases: None." in both specs; DueDateUtils "Undefined behavior: None"; all six state/due-date combinations from slide 53 and the precedence of `CLEAR`; the long-interval cases `D-42`, `D+49`, `D-52`, `D-365`; the one-calendar-month boundaries (30/31/32 days ahead, 29/30/31 days overdue); month end; the February cases; and, in the binding spec, the `null` task, the two check/uncheck sequences, the reused-view case, the CLEAR→null case and the GONE→visible case.

What refinement did not change: every Tip that ended up exposing a bug, except `D+51` (R3), was already in the AI drafts. The review did not introduce the coverage that detects bug 1 or bug 2.

5. Test plan and traceability

Each row is one Tip of a final spec and the single test that implements it, in spec order (the mapping and the literals were cross-checked by script: every Tip has exactly one test, and each expected string, and each input date for unit Tips, appears in the mapped test). Results are from the local run at `26220c8`. "AI draft scenario, reworded" means the scenario was in the AI draft and only its wording changed (R4–R6).

ID	Kind	Tip (final spec)	Test	Origin	Local	Failure message
DD-01	Success	Given dueDate = null, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "".	`dueDateLabel_activeNoDueDate_returnsEmpty`	AI draft (unchanged)	PASS	
DD-02	Success	Given dueDate = 2026-10-25, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-5".	`dueDateLabel_activeDueIn5Days_returnsDMinus5`	AI draft (unchanged)	PASS	
DD-03	Success	Given dueDate = 2026-10-20, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-Day".	`dueDateLabel_activeDueToday_returnsDDay`	AI draft (unchanged)	PASS	
DD-04	Success	Given dueDate = 2026-10-15, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+5".	`dueDateLabel_activeOverdueBy5Days_returnsDPlus5`	AI draft (unchanged)	PASS	

ID	Kind	Tip (final spec)	Test	Origin	Local	Failure message
DD-05	Success	Given dueDate = 2026-10-25, today = 2026-10-20, isCompleted = true / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "CLEAR".	`dueDateLabel_completedFutureDueDate_returnsClear`	AI draft (unchanged)	PASS	
DD-06	Success	Given dueDate = 2026-10-15, today = 2026-10-20, isCompleted = true / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "CLEAR".	`dueDateLabel_completedOverdueDueDate_returnsClear`	AI draft (unchanged)	PASS	
DD-07	Success	Given dueDate = null, today = 2026-10-20, isCompleted = true / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "CLEAR".	`dueDateLabel_completedNoDueDate_returnsClear`	AI draft (unchanged)	PASS	
DD-08	Success	Given dueDate = 2026-12-01, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-42".	`dueDateLabel_activeDue42DaysAhead_returnsDMinus42`	AI draft (unchanged)	FAIL	`expected:<D-[42]> but was:<D-[11]>`
DD-09	Success	Given dueDate = 2026-09-01, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+49".	`dueDateLabel_activeOverdueBy49Days_returnsDPlus49`	AI draft (unchanged)	FAIL	`expected:<D+[4]9> but was:<D+[1]9>`
DD-10	Success	Given dueDate = 2027-02-10, today = 2026-12-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-52".	`dueDateLabel_activeDue52DaysAheadAcrossYearEnd_returnsDMinus52`	AI draft (unchanged)	FAIL	`expected:<D-[52]> but was:<D-[21]>`
DD-11	Success	Given dueDate = 2026-11-20, today = 2027-01-10, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+51".	`dueDateLabel_activeOverdueBy51DaysAcrossYearEnd_returnsDPlus51`	Added in review	FAIL	`expected:<D+[51]> but was:<D+[20]>`
DD-12	Success	Given dueDate = 2027-10-20, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-365".	`dueDateLabel_activeDueExactlyOneYearAhead_returnsDMinus365`	AI draft (unchanged)	FAIL	`expected:<D-[365]> but was:<D-[Day]>`
DD-13	Edge	Given dueDate = 2026-10-19, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+1".	`dueDateLabel_activeDueYesterday_returnsDPlus1`	AI draft (unchanged)	PASS	
DD-14	Edge	Given dueDate = 2026-10-20, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-Day".	`dueDateLabel_activeDueOnTodayBoundary_returnsDDay`	AI draft (unchanged)	PASS	
DD-15	Edge	Given dueDate = 2026-10-21, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-1".	`dueDateLabel_activeDueTomorrow_returnsDMinus1`	AI draft (unchanged)	PASS	
DD-16	Edge	Given dueDate = 2026-11-19, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-30".	`dueDateLabel_activeDue30DaysAhead_returnsDMinus30`	AI draft (unchanged)	PASS	

ID	Kind	Tip (final spec)	Test	Origin	Local	Failure message
DD-17	Edge	Given dueDate = 2026-11-20, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-31".	`dueDateLabel_activeDue31DaysAhead_returnsDMinus31`	AI draft (unchanged)	FAIL	`expected:<D-[31]> but was:<D-[Day]>`
DD-18	Edge	Given dueDate = 2026-11-21, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-32".	`dueDateLabel_activeDue32DaysAhead_returnsDMinus32`	AI draft (unchanged)	FAIL	`expected:<D-[32]> but was:<D-[1]>`
DD-19	Edge	Given dueDate = 2026-09-21, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+29".	`dueDateLabel_activeOverdueBy29Days_returnsDPlus29`	AI draft (unchanged)	PASS	
DD-20	Edge	Given dueDate = 2026-09-20, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+30".	`dueDateLabel_activeOverdueBy30Days_returnsDPlus30`	AI draft (unchanged)	FAIL	`expected:<D[+30]> but was:<D[-Day]>`
DD-21	Edge	Given dueDate = 2026-09-19, today = 2026-10-20, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+31".	`dueDateLabel_activeOverdueBy31Days_returnsDPlus31`	AI draft (unchanged)	FAIL	`expected:<D+[3]1> but was:<D+[]1>`
DD-22	Edge	Given dueDate = 2026-10-30, today = 2026-10-31, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+1".	`dueDateLabel_dueDayBeforeMonthEnd_returnsDPlus1`	AI draft (unchanged)	PASS	
DD-23	Edge	Given dueDate = 2026-10-31, today = 2026-10-31, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-Day".	`dueDateLabel_dueOnMonthEnd_returnsDDay`	AI draft (unchanged)	PASS	
DD-24	Edge	Given dueDate = 2026-11-01, today = 2026-10-31, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-1".	`dueDateLabel_dueDayAfterMonthEnd_returnsDMinus1`	AI draft (unchanged)	PASS	
DD-25	Edge	Given dueDate = 2026-12-30, today = 2026-12-31, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+1".	`dueDateLabel_dueDayBeforeYearEnd_returnsDPlus1`	Added in review	PASS	
DD-26	Edge	Given dueDate = 2026-12-31, today = 2026-12-31, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-Day".	`dueDateLabel_dueOnYearEnd_returnsDDay`	Added in review	PASS	
DD-27	Edge	Given dueDate = 2027-01-01, today = 2026-12-31, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-1".	`dueDateLabel_dueNewYearsDayFromYearEnd_returnsDMinus1`	AI draft (unchanged)	PASS	
DD-28	Edge	Given dueDate = 2026-12-31, today = 2027-01-01, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D+1".	`dueDateLabel_dueYearEndFromNewYearsDay_returnsDPlus1`	AI draft (unchanged)	PASS	

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DD-29	Edge	Given dueDate = 2028-02-29, today = 2028-02-28, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-1".	`dueDateLabel_dueLeapDayFromFeb28_returnsDMinus1`	AI draft (unchanged)	PASS	
DD-30	Edge	Given dueDate = 2028-03-01, today = 2028-02-28, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-2".	`dueDateLabel_dueMarch1FromFeb28InLeapYear_returnsDMinus2`	AI draft (unchanged)	PASS	
DD-31	Edge	Given dueDate = 2027-03-01, today = 2027-02-28, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-1".	`dueDateLabel_dueMarch1FromFeb28InCommonYear_returnsDMinus1`	AI draft (unchanged)	PASS	
DD-32	Edge	Given dueDate = 2028-03-01, today = 2028-02-01, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-29".	`dueDateLabel_wholeFebruaryInLeapYear_returnsDMinus29`	AI draft (unchanged)	FAIL	`expected:<D-[29]> but was:<D-[Day]>`
DD-33	Edge	Given dueDate = 2027-03-01, today = 2027-02-01, isCompleted = false / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "D-28".	`dueDateLabel_wholeFebruaryInCommonYear_returnsDMinus28`	AI draft (unchanged)	FAIL	`expected:<D-[28]> but was:<D-[Day]>`
DD-34	Edge	Given dueDate = 2026-10-20, today = 2026-10-20, isCompleted = true / When dueDateLabel(dueDate, today, isCompleted) is called / Then it returns "CLEAR".	`dueDateLabel_completedDueToday_returnsClear`	AI draft (unchanged)	PASS	
TLB-01	Success	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(3)) (e.g. 2026-10-20 → 2026-10-23) / When setDueDateLabel(textView, task) is called / Then textView.text is "D-3" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeDueIn3Days_showsDMinus3`	AI draft scenario, reworded in review	PASS	
TLB-02	Success	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(0)) (e.g. 2026-10-20 → 2026-10-20) / When setDueDateLabel(textView, task) is called / Then textView.text is "D-Day" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeDueToday_showsDDay`	AI draft scenario, reworded in review	PASS	
TLB-03	Success	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(-3)) (e.g. 2026-10-20 → 2026-10-17) / When setDueDateLabel(textView, task) is called / Then textView.text is "D+3" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeOverdueBy3Days_showsDPlus3`	AI draft scenario, reworded in review	PASS	
TLB-04	Success	Given a new TextView and Task(title = "Task", isCompleted = true, dueDate = today.plusDays(3)) (e.g. 2026-10-20 → 2026-10-23) / When setDueDateLabel(textView, task) is called / Then textView.text is "CLEAR" and textView.visibility is View.VISIBLE.	`setDueDateLabel_completedWithDueDate_showsClear`	AI draft scenario, reworded in review	PASS	

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TLB-05	Success	Given a new TextView and Task(title = "Task", isCompleted = true, dueDate = null) / When setDueDateLabel(textView, task) is called / Then textView.text is "CLEAR" and textView.visibility is View.VISIBLE.	`setDueDateLabel_completedNoDueDate_showsClear`	AI draft (unchanged)	PASS	
TLB-06	Success	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = null) / When setDueDateLabel(textView, task) is called / Then textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE is true.	`setDueDateLabel_activeNoDueDate_showsNoLabel`	AI draft scenario, reworded in review	PASS	
TLB-07	Success	Given a new TextView / When setDueDateLabel(textView, null) is called / Then textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE is true.	`setDueDateLabel_nullTask_showsNoLabel`	AI draft scenario, reworded in review	PASS	
TLB-08	Success	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(42)) (e.g. 2026-10-20 → 2026-12-01) / When setDueDateLabel(textView, task) is called / Then textView.text is "D-42" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeDue42DaysAhead_showsDMinus42`	AI draft scenario, reworded in review	FAIL	`text (dueDate=2026-12-01) expected:<D-[42]> but was:<D-[11]>`
TLB-09	Edge	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(-1)) (e.g. 2026-10-20 → 2026-10-19) / When setDueDateLabel(textView, task) is called / Then textView.text is "D+1" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeDueYesterday_showsDPlus1`	AI draft scenario, reworded in review	PASS	
TLB-10	Edge	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(0)) (e.g. 2026-10-20 → 2026-10-20) / When setDueDateLabel(textView, task) is called / Then textView.text is "D-Day" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeDueOnTodayBoundary_showsDDay`	AI draft scenario, reworded in review	PASS	
TLB-11	Edge	Given a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(1)) (e.g. 2026-10-20 → 2026-10-21) / When setDueDateLabel(textView, task) is called / Then textView.text is "D-1" and textView.visibility is View.VISIBLE.	`setDueDateLabel_activeDueTomorrow_showsDMinus1`	AI draft scenario, reworded in review	PASS	
TLB-12	Edge	Given one TextView and Task(title = "Task", isCompleted = false, dueDate = null, id = "t1") / When setDueDateLabel is called with that task, then with its copy(isCompleted = true), then with its copy(isCompleted = false) / Then after the second call textView.text is "CLEAR" and textView.visibility is View.VISIBLE, and after the third call textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE is true.	`setDueDateLabel_noDueDateTaskCheckedThenUnchecked_showsClearThenNoLabel`	AI draft scenario, reworded in review	FAIL	`expected no label but saw Shown(text=CLEAR, visibility=0) (after uncheck)`

ID	Kind	Tip (final spec)	Test	Origin	Local	Failure message
TLB-13	Edge	Given one TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(3), id = "t1") (e.g. 2026-10-20 → 2026-10-23) / When setDueDateLabel is called with that task, then with its copy(isCompleted = true), then with its copy(isCompleted = false) / Then textView.text is "D-3", then "CLEAR", then "D-3", and textView.visibility is View.VISIBLE after each call.	`setDueDateLabel_dueDateTaskCheckedThenUnchecked_showsDMinus3ThenClearThenDMinus3`	AI draft scenario, reworded in review	PASS	
TLB-14	Edge	Given one TextView already bound to Task(title = "A", isCompleted = false, dueDate = today.plusDays(3)) (e.g. 2026-10-20 → 2026-10-23) so it shows "D-3" / When setDueDateLabel(textView, Task(title = "B", isCompleted = false, dueDate = null)) is called / Then textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE is true.	`setDueDateLabel_labelledViewReusedForTaskWithoutLabel_showsNoLabel`	AI draft scenario, reworded in review	FAIL	`expected no label but saw Shown(text=D-3, visibility=0) (after binding B)`
TLB-15	Edge	Given one TextView already bound to Task(title = "Task", isCompleted = true, dueDate = null) so it shows "CLEAR" / When setDueDateLabel(textView, null) is called / Then textView.text.toString().isEmpty() || textView.visibility != View.VISIBLE is true.	`setDueDateLabel_clearViewBoundToNull_showsNoLabel`	AI draft scenario, reworded in review	FAIL	`expected no label but saw Shown(text=CLEAR, visibility=0) (after binding null)`
TLB-16	Edge	Given a new TextView with visibility View.GONE and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(3)) (e.g. 2026-10-20 → 2026-10-23) / When setDueDateLabel(textView, task) is called / Then textView.text is "D-3" and textView.visibility is View.VISIBLE.	`setDueDateLabel_goneViewWithLabel_showsDMinus3Visible`	AI draft scenario, reworded in review	PASS	
R7	Supplementary	Task list with one active task without a due date; click the checkbox (expect CLEAR, VISIBLE), click it again (expect no label). Maps to TLB-12.	`tasksList_checkThenUncheckTaskWithoutDueDate_showsClearThenNoLabel`	R7	FAIL	`expected no label after uncheck, saw 'CLEAR' visibility=0`

6. Test execution and results

6.1 Environment

- Host: Windows 11; Gradle 8.13 wrapper; JDK 21.0.11 (JetBrains Runtime) (set only for the build commands because the configured `JAVA_HOME` pointed to a missing JDK 17); AGP 8.12.3; Kotlin 2.0.21.
- Device: Android emulator AVD `Pixel_6`, image `sdk_gphone16k_x86_64`, Android 17 (API 37), cold boot with 3 GB RAM and 4 cores, animations disabled, timezone GMT. Device reference date during the recorded runs: **2026-10-20 (GMT)**.
- Commands: `.\gradlew.bat testDebugUnitTest` and `.\gradlew.bat connectedDebugAndroidTest` (run as `./gradlew` in Git Bash).

6.2 Counts (local, commit `26220c8`)

Suite	Executed	Passed	Failed	Errors	Skipped
testDebugUnitTest — all	35	24	11	0	0

Suite	Executed	Passed	Failed	Errors	Skipped
DueDateUtilsTest (new)	34	23	11	0	0
connectedDebugAndroidTest — all	18	13	5	0	0
TasksListBindingsTest (new)	16	12	4	0	0
TasksFragmentDueDateTest (new, supplementary)	1	0	1	0	0
connectedDebugAndroidTest — all, API 30 emulator (commit 75ee160)	18	13	5	0	0

The API 30 row is a **local** run on a temporary emulator (system image `android-30/google_apis/x86`, Android 11), made because CI could not run (§6.4) and CI uses API level 30. Its commit `75ee160` is an empty commit on top of the evidence commit, so the code is identical to `26220c8`. It failed exactly the same 5 tests with the same messages (`D-11` for the +42-day fixture, reference date 2026-10-20, attempt 1; label view reused: true). Evidence: `docs/test-evidence/local-api30-75ee160/`.

The remaining executed tests are the course placeholders `StatisticsUtilsTest.getActiveAndCompletedStats_two_return50f50f` and `TaskDetailFragmentTest.activeTaskDetails_DisplayedInUi`. They contain no assertions, so their passing says nothing about the due-date feature.

The counts above are from a run on a clean checkout of `26220c8`. An earlier local run with byte-identical test and spec files (started before that commit was created) gave the same results: the same 11 unit and 5 instrumented failures. Every failure is a JUnit assertion failure (`ComparisonFailure` / `AssertionError`). In failure messages, `visibility=0` is `View.VISIBLE`. Raw evidence is in the repository under `docs/test-evidence/local-26220c8/` (JUnit XML, Gradle logs, filtered logcat).

6.3 Recorded fixture dates and the screen test

- Every date-dependent binding test ran on its first attempt, with reference date 2026-10-20; no date change was detected, so no attempt was discarded. Recorded due dates: +3 → 2026-10-23, −3 → 2026-10-17, −1 → 2026-10-19, 0 → 2026-10-20, +1 → 2026-10-21, +42 → 2026-12-01 (observed `D-11`).
- The screen test waits, with a 5-second bound, until the first row has been re-bound to the task with the new completion state. This is a data-side condition, because `ListAdapter` diffs on a background thread that Espresso does not track. Only then does it assert the label, so a wrong label is a genuine assertion failure, not a timeout. It does not use fixed sleeps and does not block the UI thread. It also checked object identity: the label `TextView` was the **same instance** before checking, after checking and after unchecking (logcat: "label view reused across check/uncheck: true"). This is an observation on this device, not an inference from the disabled change animations.

6.4 Environment and infrastructure issues (not test results)

- First emulator boot: the guest was overloaded (load average ≈ 588; `pm/settings` hung). The first `connectedDebugAndroidTest` attempt therefore never started; no test executed and no result was produced or discarded. A cold boot with more memory and cores fixed it.
- CI: The student enabled GitHub Actions on the fork; the PR then triggered run 39366136282 on commit 75ee160. GitHub refused to start both jobs: "The job was not started because your account is locked due to a billing issue." No runner was assigned, no step ran, and no JUnit report was produced. This is a GitHub account issue, not a repository, build or test failure, and it says nothing about the tests. CI evidence for this assignment is therefore missing; the conclusions rest on the local runs.

7. Comparison with the actual bugs

Bug	Spec condition	Failing tests	Expected	Actual	Status
1. Wrong D-day count	DueDateUtils: "N counts every calendar day" (DD-08…12), one-month boundaries (DD-17/18, DD-20/21), whole February (DD-32/33); TasksListBindings: 42-day case (TLB-08)	11 unit tests + `setDueDateLabel_activeDue42DaysAhead_showsDMinus42`	e.g. `D-42`, `D-31`, `D+30`, `D-365`	`D-11`, `D-Day`, `D-Day`, `D-Day`	Detected (local; CI: not executed — account billing lock)
2. Stale label not cleared	TasksListBindings: "shows no label" in TLB-12 (check → uncheck, no due date), TLB-14 (reused view), TLB-15 (CLEAR → null)	3 direct-call binding tests + supplementary screen test	no label (`text.isEmpty() || visibility != VISIBLE`)	visible `CLEAR` (TLB-12, TLB-15, screen), visible `D-3` (TLB-14)	Detected (local; CI: not executed — account billing lock)

7.1 Bug 1 — the count uses only the day part

Intended: N is the total number of calendar days between `today` and `dueDate` (slide 53; PR: "N counts calendar days"). **Observed cause:** `DueDateUtils.kt` computes `Period.between(today, dueDate).days`, which is only the days component of a years/months/days period. Representative test: `DueDateUtilsTest.dueDateLabel_activeDue42DaysAhead_returnsDMinus42`, with dueDate 2026-12-01, today 2026-10-20 and isCompleted false. It expects `D-42` and gets `D-11`, because the period is 1 month 11 days.

ID	Test	Input	Assertion failure
DD-08	`dueDateLabel_activeDue42DaysAhead_returnsDMinus42`	dueDate = 2026-12-01, today = 2026-10-20, isCompleted = false	`expected:<D-[42]> but was:<D-[11]>`
DD-09	`dueDateLabel_activeOverdueBy49Days_returnsDPlus49`	dueDate = 2026-09-01, today = 2026-10-20, isCompleted = false	`expected:<D+[4]9> but was:<D+[1]9>`
DD-10	`dueDateLabel_activeDue52DaysAheadAcrossYearEnd_returnsDMinus52`	dueDate = 2027-02-10, today = 2026-12-20, isCompleted = false	`expected:<D-[52]> but was:<D-[21]>`
DD-11	`dueDateLabel_activeOverdueBy51DaysAcrossYearEnd_returnsDPlus51`	dueDate = 2026-11-20, today = 2027-01-10, isCompleted = false	`expected:<D+[51]> but was:<D+[20]>`
DD-12	`dueDateLabel_activeDueExactlyOneYearAhead_returnsDMinus365`	dueDate = 2027-10-20, today = 2026-10-20, isCompleted = false	`expected:<D-[365]> but was:<D-[Day]>`
DD-17	`dueDateLabel_activeDue31DaysAhead_returnsDMinus31`	dueDate = 2026-11-20, today = 2026-10-20, isCompleted = false	`expected:<D-[31]> but was:<D-[Day]>`
DD-18	`dueDateLabel_activeDue32DaysAhead_returnsDMinus32`	dueDate = 2026-11-21, today = 2026-10-20, isCompleted = false	`expected:<D-[32]> but was:<D-[1]>`
DD-20	`dueDateLabel_activeOverdueBy30Days_returnsDPlus30`	dueDate = 2026-09-20, today = 2026-10-20, isCompleted = false	`expected:<D[+30]> but was:<D[-Day]>`

ID	Test	Input	Assertion failure
DD-21	`dueDateLabel_activeOverdueBy31Days_returnsDPlus31`	dueDate = 2026-09-19, today = 2026-10-20, isCompleted = false	`expected:<D+[3]1> but was:<D+[]1>`
DD-32	`dueDateLabel_wholeFebruaryInLeapYear_returnsDMinus29`	dueDate = 2028-03-01, today = 2028-02-01, isCompleted = false	`expected:<D-[29]> but was:<D-[Day]>`
DD-33	`dueDateLabel_wholeFebruaryInCommonYear_returnsDMinus28`	dueDate = 2027-03-01, today = 2027-02-01, isCompleted = false	`expected:<D-[28]> but was:<D-[Day]>`
TLB-08	`setDueDateLabel_activeDue42DaysAhead_showsDMinus42`	a new TextView and Task(title = "Task", isCompleted = false, dueDate = today.plusDays(42)) (e.g. 2026-10-20 → 2026-12-01)	`text (dueDate=2026-12-01) expected:<D-[42]> but was:<D-[11]>`

Why these tests, and why others pass. A test can expose this bug only if the interval has a non-zero months or years component. That is why the 30-day case passes (period 30 days) while the 31-day case fails (period exactly 1 month → `D-Day`), and the 32-day case fails as `D-1`. Short calendar-boundary cases (month end, year end, leap day, ±1 day) pass on the buggy code: they check boundary correctness but cannot tell total days from the day component. The binding-level 42-day test exposes the bug on any run date, because 42 days always contain at least one whole month.

Origin of coverage: 10 of the 11 failing unit Tips and the binding 42-day Tip were in the AI drafts; R3 added one more failing case (`D+51`). Refinement was not necessary to detect bug 1.

7.2 Bug 2 — a stale label is not cleared

Intended: an active task without a due date shows no label, and the label updates when the task is checked or unchecked (slide 53). **Observed cause:** `setDueDateLabel` writes text and `VISIBLE` only `if (label.isNotEmpty())`. For an empty label it does nothing, so whatever an earlier binding showed in the same view stays. Representative test: `TasksListBindingsTest.setDueDateLabel_noDueDateTaskCheckedThenUnchecked_showsClearThenNoLabel`. On one `TextView` it binds Task("Task", isCompleted=false, dueDate=null, id="t1"), then `copy(isCompleted=true)`, then `copy(isCompleted=false)`. After the check it shows `CLEAR` (pass); after the uncheck it expects no label, but the view still shows `CLEAR` and is VISIBLE.

ID	Test	Assertion failure
TLB-12	`setDueDateLabel_noDueDateTaskCheckedThenUnchecked_showsClearThenNoLabel`	`expected no label but saw Shown(text=CLEAR, visibility=0) (after uncheck)`
TLB-14	`setDueDateLabel_labelledViewReusedForTaskWithoutLabel_showsNoLabel`	`expected no label but saw Shown(text=D-3, visibility=0) (after binding B)`
TLB-15	`setDueDateLabel_clearViewBoundToNull_showsNoLabel`	`expected no label but saw Shown(text=CLEAR, visibility=0) (after binding null)`
R7	`tasksList_checkThenUncheckTaskWithoutDueDate_showsClearThenNoLabel`	`expected no label after uncheck, saw 'CLEAR' visibility=0`

Why these tests, and why others pass. The defect is visible only when the *same* view already displays a label. Tests on a new `TextView` (`activeNoDueDate`, `nullTask`) pass, because a new view is already empty: they check the stateless contract and cannot reveal staleness. The check/uncheck sequence *with* a due date passes, because the new non-empty `D-3` overwrites `CLEAR`. The screen test reproduces the user-visible symptom described on slide 68 through the real checkbox, and it confirmed that the row's label view instance was reused.

Origin of coverage: all three failing direct-call scenarios were in the AI draft. Review made the "no label" oracle explicit and exact (R4) and added the screen-level evidence (R7). Refinement was not necessary to detect bug 2.

8. Limitations

- CI could not execute any test because the GitHub account is locked for billing (§6.4), so there is no CI evidence. Instrumented tests were run locally on API 37 and API 30 emulators; the CI emulator (x86_64, API 30, Ubuntu runner) was not exercised.
- The binding tests depend on the device clock. The race handling is designed and implemented, but no midnight crossing happened in the recorded runs, so the retry path itself was not exercised.

- No mutation-testing campaign or coverage target was run; neither is a submission requirement.
- The production defects are deliberately left unfixed, as the assignment requires. The PR must not be merged while these tests fail.

9. Integrity checks

- `DueDateUtils-AI.spec.md` SHA-256 `6b72877716a5a4ec2306641e70f5a9f8c56404c97ea522a686bd864588baefc7`; `TasksListBindings-AI.spec.md` SHA-256 `6ed86b1743bfc176b2e1c3ac2f965176d646b1321416b953a260818b1ee99297`. These are identical to the hashes recorded when the drafts were created (commit `e517122`).
- `git diff 7e091a7 -- app/src/main app/build.gradle.kts build.gradle.kts settings.gradle.kts gradle .github` is empty: no production code, build or CI configuration was changed.
- Commits on `feature/due-date` added for this assignment: `e517122` (AI specs), `df03528` (refined specs), `26220c8` (tests), 265f66a (local evidence), 75ee160 (empty commit to trigger CI after Actions was enabled), followed by one documentation-only commit adding the API 30 evidence and this report.